Gradient descent prunes PING via Dale's-law clamping

exp049 · 9 June 2026 · Draft

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

Abstract

exp025 freezes the recurrent inhibitory weights W^{EI}, W^{IE} at biophysical values. Unfreeze them and Adam does *not* rediscover PING: from any initialisation — canonical, zero, or 10% of canonical — training prunes most W^{EI} entries below zero, the Dale's-law forward-pass clamp turns them into structural zeros, and the network collapses to dense E firing with I silent, at ≈ 88% accuracy (≈ 2 pp *above* the frozen PING control). PING is a structural prior the freeze *imposes*, not one gradient descent recovers on its own.

Method

Architecture. $N_E = 1024$ excitatory, $N_I = 256$ inhibitory, mem-mean readout, Dale's law enforced. Hyperparameters match the exp025 PING baseline: Adam at lr 4×10^{-4}, batch 256, surrogate slope 1, $W_{\text{in}} \sim \mathcal{N}(1.2, 0.12)$ at 95% sparsity, gradient norm clipped to 1.0, $\Delta t = 0.1$ ms, $T = 200$ ms, no firing-rate regulariser.

Sweep. Four conditions × three seeds (42, 43, 44), medium tier (1600 train / 400 test MNIST, 100 epochs). Only the initial (W^{EI}, W^{IE}) and the trainable-or-not flag vary:

Condition	W^{EI}, W^{IE} init	Trainable?
frozen_ping (control)	canonical biophysical	no
trainable_ping_init	canonical biophysical	**yes**
trainable_zero_init	0 (COBA-equivalent start)	**yes**
trainable_small_init	0.1 × canonical	**yes**

Canonical biophysical means $W^{EI} \sim \mathcal{N}(1.0, 0.1)$ µS, $W^{IE} \sim \mathcal{N}(2.0, 0.2)$ µS at $N_I = 256$, fan-in-normalised — so the trainer reports per-edge means of ≈ 0.0010 and ≈ 0.0078. "PING is on" means the inhibitory loop is alive: I fires at a healthy rate and paces a gamma rhythm; "the loop is gone" means I is silenced and E fires densely (plain COBA). The firing rates are the cleanest read of which regime a trained network lands in.

Results

The whole sweep collapses to one picture. In the (E firing rate, I firing rate) plane, a network "found PING" if it sits in the low-E / high-I corner, and "lost the loop" if it sits in the dense-E / silent-I corner.

pending re-run with new canonical data

Figure 1: Each point is one trained network (4 conditions $\times$ 3 seeds). The **frozen control** (grey) sits in the PING corner — E $\approx$ 8 Hz, I $\approx$ 38 Hz, the inhibitory loop alive. **Every trainable condition** — started at canonical PING, at zero, or at 0.1$\times$ canonical — collapses to the same dense-E / silent-I corner (E $\approx$ 45 Hz, I $\approx$ 0), at $\approx$ 88% accuracy, $\approx$ 2 pp *above* the control. Where the loop *starts* makes no difference; only whether it is allowed to train. Gradient descent never reaches PING, and the task slightly prefers it gone.

Training curves — the loop pruned, epoch by epoch

pending re-run with new canonical data

Figure 2: Per-epoch metrics from the runs themselves, coloured by init — PING init (black), zero init (red), small-seed init (amber), frozen-PING control (grey dashed); 5% of MNIST, 100 epochs, 3 seeds each. **Accuracy** — both reach $\approx$88–90% and track each other; pruning the loop costs nothing (the trainable runs even edge slightly above). **E rate** — the trainable runs climb to $\approx$25–56 Hz as the loop releases the excitatory population, while the frozen control stays gated near 8 Hz. **I rate** — every trainable run drops to $\approx$0 within a few epochs (the inhibitory loop is pruned), while the frozen control's I rate *rises* to $\approx$45 Hz as its readout trains. **Pingness** — the exp054 lobe–trough contrast: the frozen control climbs to $\approx$0.95 (the rhythm sharpens) while every trainable init collapses to $\approx$0.1–0.2. The residual floor is the metric's known low-rate inflation under shared input (exp054), not a surviving rhythm — the frozen-vs-trainable gap is the signal. The loop dies early, from every start, with no accuracy penalty.

Phase portrait — there is only one attractor under training

The four per-epoch panels above tell the story but separate the two state variables that matter. Putting *E rate* and *pingness* on perpendicular axes shows the trajectories of training directly, and the geometry rules out a misreading: it is *not* the case that PING and COBA are two attractors of the same dynamics, with the architecture deciding the basin. Under training there is **one** attractor — the COBA corner — and the PING state only persists because freezing the loop zeroes its gradients.

pending re-run with new canonical data

Figure 3: Each trainable trajectory is the per-epoch mean across 3 seeds, alpha-ramped along the epoch axis so the direction of flow reads off the line itself; open markers are epoch 1 (the first logged epoch), filled markers are epoch 100. **Frozen PING (grey)** — sits in the upper PING basin from start to finish at pingness $\approx 0.86 \to 0.94$; drifts a little in E rate ($\approx 4 \to 10$ Hz) as the feedforward and readout weights train, but the loop weights themselves can't move by construction. **Trainable PING init (black)** — was initialised with canonical biophysical loop weights, but by the time the first metric is logged (after epoch 1) the loop has already been pruned: pingness ≈ 0.17 from the start. The trajectory then walks rightward along the COBA floor with the others. **Trainable zero init (red)** and **small-seed init (amber)** — start at the floor and stay there, E rate climbing as the feedforward layer learns the task. Every legend entry lands at 87–89% accuracy, so the move costs nothing. The reading is sharper than the time-series panels suggest: the empty band between pingness ≈ 0.2 and ≈ 0.85 is the *whole story* — gradient descent flies through it within one epoch and the metric never catches it mid-flight. There is no separatrix between PING and COBA under training because there is no slow trajectory through the gap. The freeze isn't preventing slow erosion; it's preventing instant collapse.

Accuracy–rate trajectory — same destination, different spike economy

Figure 3 puts pingness on its own axis. Putting pingness on the *colour* axis instead and replaying training as (E rate, accuracy) trajectories — the exp025 accuracy–rate frontier given a time axis — gives the third reading of the same data: every condition reaches the same final test accuracy, but along very different routes, and only one of them is doing PING while it gets there.

pending re-run with new canonical data

Figure 4: Each trajectory is the per-epoch mean across 3 seeds; per-segment colour is that epoch's pingness on a viridis $0 \to 1$ scale (colourbar at right). Open markers are epoch 1, filled markers are epoch 100. The frontier *with time* tells three things at once. **Same destination** — all four conditions land at $\approx$ 87–89% accuracy by epoch 100, regardless of init or whether the loop trains. **Different routes** — the frozen control climbs accuracy almost vertically with little change in E rate ($\approx 4 \to 10$ Hz); the trainable conditions all bend right and up, the biggest one (zero init) reaching the final accuracy at E $\approx$ 57 Hz, $\approx 6\times$ the frozen control's rate. **Only one of them is still PING** — the frozen control's line is bright yellow because pingness stays high ($\approx$ 0.9) the whole way; every trainable line is dark purple because pingness was already pruned to $\approx$ 0.1 by the first logged epoch. Reading the colour bar: most of pingness 0.2–0.85 is unused space, because no trajectory crosses it slowly enough to be logged. Reading the geometry: this is the manuscript's accuracy–rate frontier (exp025, §2.3) animated — and the architecture is the only thing keeping a trained network on the sparse, rhythmic side of it. The dashed divider at $\approx$ 17 Hz marks the two basins explicitly: frozen PING holds the left, every trainable run ends in the right, at the same accuracy.